Worksheet - 1, 2 Date: 16-08-2010

1. Key: 1
2. Key: 1 Hint: AB + BC = AC (or) ΔABC
3. Key: 1
4. Key: 1
5. Key: 2
6. Key: 1, 3
7. Key: 1
8. Key: 1
9. Key: 3
10. Key: 4
11. Key: 4,4,2,(1,3)
12. Key: 3
13. Key: 2 A point P divides the straight line joining two points $A(x_1, y_1)$ and $B(x_2, y_2)$ internally in the ratio m : n is $\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$

    Since P divides the line segment joining points A (0,0) and B(9,12) internally in 1 : 2 ratio

    $\therefore \quad x = \frac{9 \times 1 + 0 \times 2}{3} = \frac{9}{3} = 3, \ y = \frac{12 \times 1 + 2 \times 0}{3} = \frac{12}{3} = 4$

    The coordinates of point P are (3, 4).
14. Key: 1,4 1) Is correct since distance of (a,b) from B and C are equal

    2) Is false since $AB^2 \neq AC^2$

    3) Is incorrect since $AB^2 \neq AC^2$

    4) Is correct since

    $AB^2 = (1-a-b)^2 + \left(\frac{b}{a} - b + a\right)^2$ and $AC^2 = (1-a+b)^2 + \left(\frac{b}{a} - a - b\right)^2$ Note that $AB^2 = AC^2$.
15. Required point = $\left\{\frac{b(a+b)+a(a-b)}{a+b}, \frac{b(a-b)+a(a+b)}{a+b}\right\} = \left\{\frac{a^2+b^2}{a+b}, \frac{a^2+2ab-b^2}{a+b}\right\}$
16. Key: 1 Let P is the point on x - axis which divides the line segment joining points in the ratio $\lambda : 1$, then, $x = \frac{-5\lambda + 3}{\lambda + 1}, y = \frac{6\lambda - 4}{\lambda + 1}$

    But for any point on x axis, its y- coordinate = 0 $\Rightarrow \frac{6\lambda - 4}{\lambda + 1} = 0 \Rightarrow \lambda = \frac{2}{3}$ (OR)

    The x - axis divides the line segment joining the points (x_1, y_1) and (x_2, y_2) in the ratio $-y_1 : y_2 = -(-4) : 6 = 4 : 6 = 2 : 3$

17. Key: 1

18. Key: 1 Given AP = PQ = QB ....(1)

Let R be the midpoint of PQ such that PR = RQ ....(2)

Now AR = AP + PR AR = QB + QR = BR [$\because$ from (1) and (2)]

$\Rightarrow$ R is the midpoint of AB $\Rightarrow R = \left(\frac{-2+3}{2}, \frac{5+1}{2}\right) = \left(\frac{1}{2}, 3\right)$.

19. Key: 4, 3, 1, 2 Hint: area of $\Delta ABC = \frac{1}{2}\begin{vmatrix} p+1 & 2p+1 & 2p+3 & p+1 \\ 1 & 3 & 2p & 1 \end{vmatrix} = \frac{1}{2}\left|2p^2 - 3p - 2\right|$

20. Key: 3 Let the ratio be m : n. P (x, y) = $(x_1 + t(x_2 - x_1), y_1 + t(y_2 - y_1))$ and $A(x_1, y_1)$, $B(x_2, y_2)$

Now, $P(x_1 + t(x_2 - x_1), y_1 + t(y_2 - y_1)) = \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$

$\therefore x_1 + t(x_2 - x_1) = \frac{mx_2 + nx_1}{m+n} \Rightarrow x_1(m+n) + t(x_2 - x_1)(m+n) = mx_2 + nx_1$

$\Rightarrow mx_1 + nx_1 + t(x_2 - x_1)(m+n) = mx_2 + nx_1 \Rightarrow t(m+n)(x_2 - x_1) = mx_2 - mx_1$

$\Rightarrow t(m+n)(x_2 - x_1) = m(x_2 - x_1) \Rightarrow m = mt + nt \Rightarrow nt = m(1-t) \qquad \therefore \frac{m}{n} = \frac{t}{1-t}$

21. Key: 3 The coordinates of the point of division are

$$\left(x_1 + t(x_2 - x_1), y_1 + t(y_2 - y_1)\right) = \left(\frac{(1-t)x_1 + tx_2}{(1-t)+t}, \frac{(1-t)y_1 + ty_2}{(1-t)+t}\right)$$

Clearly, these are the coordinates of the point dividing the join of (x_1, y_1) and (x_2, y_2) in the ratio t : (1 – t)

For the internal division, we must have $t > 0$ and $1 - t > 0 \Rightarrow 0 < t < 1$.

Worksheet - 3 Date: 17-08-2010

1. The point to which the origin has to be shifted to eliminate x and y terms in the equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2}\right)$.

$\frac{hf - bg}{ab - h^2} = \frac{(3/2)(1/2) + 2(-7/2)}{2(-2) - (3/2)^2} = 1; \frac{gh - af}{ab - h^2} = \frac{(-7/2)(3/2) - 2(1/2)}{2(-2) - (3/2)^2} = 1$

2. Key: 1 $\left(\frac{-f}{h}, \frac{-g}{h}\right) = (-1, -1)$

3. The point to which the origin has to be shifted to eliminate x and y terms in the equation $ax^2 + by^2 + 2gx + 2fy + c = 0$ is $\left(\frac{-g}{a}, -\frac{f}{b}\right)$.

$$\left(\frac{-g}{a}, -\frac{f}{b}\right) = \left(-\frac{1}{1}, +\frac{2}{1}\right) = (-1, +2)$$

4. Key : 2,4

Sol: Here a = 9, b = 3, h = $-\sqrt{3}$

angle of rotation $\theta = \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right) = \frac{1}{2}\tan^{-1}\left(\frac{-2\sqrt{3}}{9-3}\right) = \frac{1}{2}\tan^{-1}\left(\frac{-1}{\sqrt{3}}\right) = \frac{-\pi}{12}$

$\therefore \quad \theta = \frac{n\pi}{2} - \frac{\pi}{12} = \frac{5\pi}{12}$ when $= 1$

5. Key - 4 Here a = 2, h = $\frac{\sqrt{3}}{2}$, b = 3

$$\theta = \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right) = \frac{1}{2}\tan^{-1}\left(\frac{2\times \sqrt{3}/2}{2-3}\right) = \frac{1}{2}\tan^{-1}\left(-\sqrt{3}\right) \Rightarrow \theta = \frac{1}{2}\left(\frac{\pi}{2} - \frac{\pi}{6}\right) = \frac{\pi}{6}$$

6. Key: 4 Comparing the given equation with $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ we get a = 3, h = - 2, b = - 2, g = -3/2, f = - 1 required point is

$$\left(\frac{hf-bg}{ab-h^2}, \frac{gh-af}{ab-h^2}\right) = \left(\frac{2-3}{-6-4}, \frac{3+3}{-6-4}\right) = \left(\frac{1}{10}, \frac{-3}{5}\right)$$

7. Here a = 1, b = -1, h = 0, g = 1, f = 2

Required point is $\left(\frac{hf-bg}{ab-h^2}, \frac{gh-af}{ab-h^2}\right) = \left(\frac{0+1}{-1-0}, \frac{0-2}{-1-0}\right) = (-1, 2)$

8. Key: 4 Here a = 1, b = 1, h = 0, g = -a, f = -2a

reqiored point is $\left(\frac{hf-bg}{ab-h^2}, \frac{gh-af}{ab-h^2}\right) = \left(\frac{a}{1}, \frac{0+2a}{1}\right) = (a, 2a)$.

9. Key: 3,1,4,2

Sol: Let (x,y) be the new coordinates of (x,y)

a) (x,y) = (x- h, y - k) = (7-1, 5-2) = (6,3)
b) (x,y) = (-3,-1+1-2) = (-4,-1)
c) (x,y) = (0-1,5-2) = (-1,3)
d) (x,y) = (-1,-1,-2-2) = (-2,-4)

10. Key: 1

11. The first degree terms are missing.

Hence, the new origin is $\left(-\frac{f}{h}, -\frac{g}{h}\right) = \left(-\frac{1}{1/2}, \frac{1/2}{1/2}\right) = (-2, 1)$

The new equation is $xy - \frac{1}{2}(-2) + 1 - 6 = 0 \Rightarrow xy = 4$ $\therefore$ c = 4

12. New coordinates are (x –2, y + 1)
$2(x-2)^2 + 7(y+1)^2 + 8(x-2) - 14(y+1) + 15 = 0 \Rightarrow 2x^2 + 7y^2 = 0$

13. Key: 1

Sol: $(x,y) = \left((x\cos\theta - y\sin\theta), x\sin\theta + y\cos\theta\right) = \left(h.\frac{\sqrt{3}}{2} - 2\sqrt{3}.\frac{1}{2}, 4.\frac{1}{2} + 2\sqrt{3}.\frac{\sqrt{3}}{2}\right)$

$= \left(\sqrt{3}, 5\right)$

14. $\therefore$ The transformed equation is
$2(x-2)^2 + 4(x-2)(y+3) + 5(y+3)^2 - 4(x-2) - 22(y+3) + 7 = 0$
$\Rightarrow 2(x^2 - 4x + 4) + 4(xy + 3x - 2y - 6) + 5(y^2 + 6y + 9) - 4x + 8 - 22y - 66 + 7 = 0$
$\Rightarrow 2x^2 + 5y^2 - 8x + 8 + 4xy + 12x - 8y - 24 + 30y + 45 - 4x + 8 - 22y - 59 = 0$
$\Rightarrow 2x^2 + 5y^2 + 4xy - 22 = 0$

15. Key: 3

16. Key: 2

17. New coordinates are (x + 2, y –2).
The transformed equation is $2(x+2)^2 + (y-2)^2 - 8(x+2) + 4(y-2) + 1 = 0$
$2x^2 + 8x + 8 + y^2 - 4y + 4 - 8x - 16 + 4y - 8 + 1 = 0 \Rightarrow 2x^2 + y^2 - 11 = 0$

18. Key: 3,2,4,1

19. Key: 2 $y^2 - 4y = 8x - 12 \Rightarrow y^2 - 4y + 4 = 8x - 8 \Rightarrow (y-2)^2 = 8(x-1)$
If the origin is shifted to (1,2), then the equation becomes $y^2 = 8x$
But the given equation is $y^2 = 4ax \Rightarrow a = 2$

20. Let (x, y) be the original coordinates of P.

$x = \cos\alpha + 1$ and $y = \cos\beta + 1 \Rightarrow (x, y) = \left(2\cos^2\frac{\alpha}{2}, 2\cos^2\frac{\beta}{2}\right)$

$\left(\because 1 + \cos 2\theta = 2\cos^2\theta\right)$

21. Key: 2,4,1,3

Cumulative Assignment Date: 18-08-2010

1. Key: 1 The point to which the origin has to be shifted to eliminate x and y terms in the equation $ax^2 + by^2 + 2gx + 2fy + c = 0$ is $\left(\frac{-g}{a}, -\frac{f}{b}\right)$. $\left(-\frac{g}{a}, -\frac{f}{b}\right) = \left(-\frac{4}{2}, \frac{7}{7}\right) = (-2, 1)$

2. Key: 2 The point to which the origin has to be shifted to eliminate x and y terms in the equation $ax^2 + by^2 + 2gx + 2fy + c = 0$ is $\left(\frac{-g}{a}, -\frac{f}{b}\right)$.

$$\left(-\frac{g}{a},-\frac{f}{b}\right)=\left(\frac{+2}{1},\frac{-3}{1}\right)=(+2,-3)$$

3. Key: 4 $(x,y)=\left(\sqrt{2},-\sqrt{2}\right)$ and $\theta=45^0, X=x\cos\theta+y\sin\theta=\sqrt{2}\cos45^0-\sqrt{2}\sin45^0=0$

$Y=-\sqrt{2}\sin45^0-\sqrt{2}\cos45^0=-2\Rightarrow(X,Y)=(0,-2)$

4. Let (X,Y) be the new coordinates of (x,y) angle of rotation $\theta=45°$

now x = X cos45° - Y sin45° = $\frac{x-y}{\sqrt{2}}, y$ Xsin45° + Ycos45° = $\frac{x+y}{\sqrt{2}}$

The transformed equation is

$$\Rightarrow 17\left(\frac{x-y}{\sqrt{2}}\right)^2-16\left(\frac{x-y}{\sqrt{2}}\right)\left(\frac{x+y}{\sqrt{2}}\right)+17\left(\frac{x+y}{\sqrt{2}}\right)^2=225\Rightarrow 9x^2+25y^2=225$$

5. Key: 1
6. Key: 3
Hint : x = X cos60 – Y sin 60, y = X sin 60 + Y cos60

$$x=X.\tfrac{1}{2}-X\frac{\sqrt{3}}{2}, y=X.\frac{\sqrt{3}}{2}+Y.\tfrac{1}{2}\Rightarrow(x,y)=\left(\frac{x-\sqrt{3}y}{2},\frac{\sqrt{3}x+y}{2}\right)$$

substitute (x,y) in $7x^2+2\sqrt{3}\ xy+9y^2=8$

7. Key: 4
8. Key: 1 Hint: $x=x\cos\pi/4-y\sin\pi/4; y=x\sin\pi/4+y\cos\pi/4$
9. Key: $a\to2$, $b\to1$, $c\to3$, $d\to4$
10. Key: 3 $(X,Y)=\left(2,-\sqrt{3}\right)$ and $\theta=60^0, x=2\cos60-\left(-\sqrt{3}\right)\sin60=2(1/2)+\sqrt{3}\left(\sqrt{3}/2\right)=\frac{5}{2}$

$y=2\sin60+\left(-\sqrt{3}\right)\cos60=2\left(\sqrt{3}/2\right)-\sqrt{3}(1/2)=\frac{\sqrt{3}}{2}$ $\therefore (x,y)=\left(\frac{5}{2},\frac{\sqrt{3}}{2}\right)$

11. Key: 4 Let θ be the angle of rotation

The new equation is $(x\cos\theta-y\sin\theta)\sqrt{3}+x\sin\theta+y\cos\theta=2$

In this y term is absent $\Rightarrow-\sin\theta\sqrt{3}+\cos\theta=0\Rightarrow\tan\theta=1/\sqrt{3}\Rightarrow\theta=30^0$

12. Key: 1 New equation is $(x\cos\alpha-y\sin\alpha)\cos\alpha+(x\sin\alpha+y\cos\alpha)\sin\alpha=p$

$\Rightarrow x\left(\cos^2\alpha+\sin^2\alpha\right)=p\Rightarrow x=p$

13. Key: 1

14. The new coordinates are $x = X\cos 180^0 + Y\sin 180^0 = -X$

$y = -X\sin 180^0 + Y\cos 180^0 = -Y$

The new equation is $(-X)-2(-Y)+3=0 \Rightarrow X-2Y-3=0$

15. The new coordinates are $x = X\cos 90^0 + Y\sin 90^0 = Y$

$y = -X\sin 90^0 + Y\cos 90^0 = -X$

The new equation is $\frac{(Y)^2}{a^2}+\frac{(-X)^2}{b^2}=1 \Rightarrow \frac{X^2}{b^2}+\frac{Y^2}{a^2}=1$

16. Key: 4

17. $\left[x\cos(-45)-y\sin(-45)\right]^2-\left[x\sin(-45)+y\cos(-45)\right]^2=a^2$

$$\Rightarrow \left(\frac{x}{\sqrt{2}}+\frac{y}{\sqrt{2}}\right)^2-\left(\frac{-x}{\sqrt{2}}+\frac{y}{\sqrt{2}}\right)^2=a^2 \Rightarrow (x+y)^2-(y-x)^2=2a^2 \Rightarrow 4xy=2a^2 \Rightarrow 2xy=a^2$$

18. The image of $(2,-1)$ w.r. to y - axis is $(-2, -1)$ and $\theta=-45^0$

$$X=(-2)\cos(-45)+(-1)\sin(-45)=-\frac{2}{\sqrt{2}}+\frac{1}{\sqrt{2}}=-\frac{1}{\sqrt{2}}$$

$$Y=-(-2)\sin(-45)+(-1)\cos(-45)=-\frac{2}{\sqrt{2}}-\frac{1}{\sqrt{3}}=-\frac{3}{\sqrt{2}} \Rightarrow (X,Y)=\left(-\frac{1}{\sqrt{2}},-\frac{3}{\sqrt{2}}\right)$$

19. Key: $a\to 4$, $b\to 3$, $c\to 2$, $d\to 1$

Worksheet - 4, 5 Date: 19-08-2010

1. Slope of the line is tan 45° = 1.
Equation of the line passing through (1, 0) and having slope 1, is
$(y-0)=1(x-1) \Rightarrow x-y-1=0$

2. Let the equation of the line be $\frac{x}{a}+\frac{y}{b}=1$ ....(1)

Given : intercept on y-axis = 2 (intercept on x - axis) i.e. b = 2a.

So, equation (1) becomes $\frac{x}{a}+\frac{y}{2a}=1 \Rightarrow 2x+y=2a$ ....(2)

Since line (2) passes through the point (3, 4) $\therefore\ 2\times 3+4=2a \Rightarrow a=5$

Thus, the equation of the required line is $2x+y=10$

3. Slope (m) $=\frac{y_2-y_1}{x_2-x_1}=\frac{2+4}{-3-5}=-\frac{3}{4}$

4. Given equation $\frac{x}{a}+\frac{y}{b}=1 \Rightarrow bx+ay=ab \Rightarrow ay=-bx+ab \Rightarrow y=-\frac{b}{a}x+b$

It is in the form of y = mx + c $\therefore$ Slope = $\frac{-b}{a}$

5. Key: 2,2
6. Key: 3

The equation of line parallel to $ax+by+c=0$ and passing through (x_1,y_1) is $a(x-x_1)+b(y-y_1)=0$

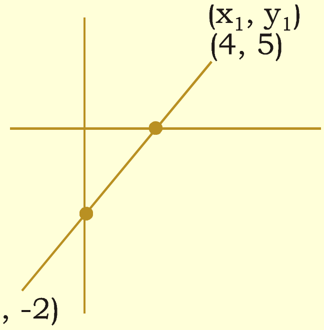
7. Key: 1
8. Key: 2
9. Key: 3
10. Key : 3,2,5,1
11. Key: 2 Hint Slope = $\frac{-8}{-4}=2$
12. m = tan60° $=\sqrt{3}$ and the y - intercept is 3

$\therefore$ The required equation is $y=\sqrt{3}x+3 \Rightarrow x\sqrt{3}-y+3=0$

13(1). Key: 1, 2

13.(2). Key: 1, 3

Let point on x axis be (x, 0) and point on y-axis O be (0, y)



$\frac{my_2+ny_1}{m+n}=0 \Rightarrow \frac{m(-2)+n(5)}{m+n}=0 \Rightarrow -2m+5n=0 \Rightarrow 2m=5n \Rightarrow \frac{m}{n}=\frac{5}{2}$

$\frac{mx_2+nx_1}{m+n}=0 \Rightarrow \frac{m(-10)+n(4)}{m+n}=0 \Rightarrow -10m+4n=0 \Rightarrow 10m=4n \Rightarrow \frac{m}{n}=\frac{2}{5}$

14. Given points are $A\left(am_1^2, 2am_1\right), B\left(am_2^2, 2am_2\right)$

Slope of $AB(m)=\frac{y_2-y_1}{x_2-x_1} \Rightarrow m=\frac{2am_2-2am_1}{am_2^2-am_1^2}=\frac{2(m_2-m_1)}{m_2^2-m_1^2}=\frac{2}{m_1+m_2}$

$\therefore$ The required equation is $y-y_1=m(x-x_1) \Rightarrow y-2am_1=\frac{2}{m_1+m_2}\left(x-am_1^2\right)$

$\Rightarrow y(m_1+m_2)-2am_1(m_1+m_2)=2x-2am_1^2$

$\therefore 2x - y(m_1 + m_2) - 2am_1^2 - 2am_1m_2 + 2am_1^2 = 0$

$\Rightarrow 2x - y(m_1 + m_2) - 2am_1m_2 = 0,$ is the required equation.

15. Given points are $A(at_1^2, 2at_1)$ and $B(at_2^2, 2at_2)$

Slope of AB = $\dfrac{2at_2 - 2at_1}{at_2^2 - at_1^2} = \dfrac{2a(t_2 - t_1)}{a(t_2 - t_1)(t_2 + t_1)} = \dfrac{2}{t_2 + t_1}$

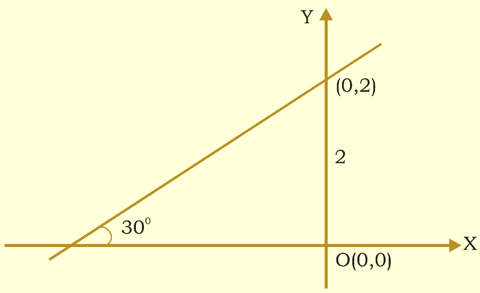
But the given line is parallel to y = x $\Rightarrow \dfrac{2}{t_2 + t_1} = 1 \Rightarrow t_1 + t_2 = 2$

16. Key: 2,3

Let the slope of the line is m, then $\dfrac{1}{2} = \left|\dfrac{m - (-2)}{1 + (-2)m}\right| = m = {}^{-3}\!/_{4}$ or ∞

Hence equation of line is $y - 3 = \dfrac{-3}{4}(x - 2)$ *and* $x = 2$

17. Given y - intercept (c) = 2 and $\theta = 30^0 \Rightarrow m = \tan\theta = \tan 30° = \dfrac{1}{\sqrt{3}}$



$\therefore$ Required equation is y = mx + c $\Rightarrow y = \dfrac{1}{\sqrt{3}}x + 2$

$\Rightarrow \sqrt{3}y = x + 2\sqrt{3} \Rightarrow x - \sqrt{3}y + 2\sqrt{3} = 0$

18. The equation is $\dfrac{y - y_1}{y_2 - y_1} = \dfrac{x - x_1}{x_2 - x_1} \Rightarrow \dfrac{y - \frac{c}{t_1}}{\frac{c}{t_2} - \frac{c}{t_1}} = \dfrac{x - ct_1}{ct_2 - ct_1} \Rightarrow \dfrac{\frac{t_1y - c}{t_1}}{\frac{c(t_1 - t_2)}{t_1t_2}} = \dfrac{x - ct_1}{c(t_2 - t_1)}$

$\Rightarrow (t_1y - c)t_2 = -(x - ct_1) \Rightarrow t_1t_2y - ct_2 = -x + ct_1$

$x + t_1t_2y = ct_1 + ct_2 \qquad x + t_1t_2y = c(t_1 + t_2)$

19. Key- $a \to 3,\ b \to 2,\ c \to 4,\ d \to 1$.

20. Given points are $A(c\cos\alpha, c\sin\alpha)$ and $B(c\cos\beta, c\sin\beta)$

Slope of AB is (m) $= \dfrac{c\sin\beta - c\sin\alpha}{c\cos\beta - c\cos\alpha} = \dfrac{\sin\beta - \sin\alpha}{\cos\beta - \cos\alpha}$

Equation of AB is $y - y_1 = m(x - x_1) \Rightarrow y - c\sin\alpha = \dfrac{\sin\beta - \sin\alpha}{\cos\beta - \cos\alpha}(x - c\cos\alpha)$

$\Rightarrow y(\cos\beta - \cos\alpha) - c\sin\alpha(\cos\beta - \cos\alpha) = (x - \cos\alpha)(\sin\beta - \sin\alpha)$

$\Rightarrow y\left[-2\sin\left(\dfrac{\beta+\alpha}{2}\right)\sin\left(\dfrac{\beta-\alpha}{2}\right)\right] - c\sin\alpha\left[-2\sin\left(\dfrac{\beta+\alpha}{2}\right)\sin\left(\dfrac{\beta-\alpha}{2}\right)\right]$

$= (x - c\cos\alpha)2\cos\left(\dfrac{\beta+\alpha}{2}\right)\sin\left(\dfrac{\beta-\alpha}{2}\right)$

$\Rightarrow -y\sin\left(\dfrac{\alpha+\beta}{2}\right) + c\sin\alpha\sin\left(\dfrac{\beta+\alpha}{2}\right) = (x - c\cos\alpha)\cos\left(\dfrac{\alpha+\beta}{2}\right)$

$\Rightarrow x\cos\left(\dfrac{\alpha+\beta}{2}\right) + y\sin\left(\dfrac{\alpha+\beta}{2}\right) = c\sin\alpha\sin\left(\dfrac{\alpha+\beta}{2}\right) + c\cos\alpha\ \cos\left(\dfrac{\beta+\alpha}{2}\right)\ = c\cos\left[\alpha - \dfrac{\alpha+\beta}{2}\right]$

$\therefore\ x\cos\left(\dfrac{\alpha+\beta}{2}\right) + y\sin\left(\dfrac{\alpha+\beta}{2}\right) = c\cos\left(\dfrac{\alpha+\beta}{2}\right)$

21. $4a^2 + 9b^2 + 12ab - c^2 = 0 \Rightarrow (2a+3b)^2 - c^2 = 0$

$\Rightarrow (2a+3b+c)(2a+3b-c) = 0 \Rightarrow 2a+3b+c = 0$ or $2a+3b-c = 0$

$\Rightarrow 2a+3b+c = 0$ or $-2a-3b+c = 0$, this shows that the line ax + by + c = 0 passes through (2, 3), (–2, –3)

22. Given line is $\left(\dfrac{1}{a}+\dfrac{k}{b}\right)x + \left(\dfrac{1}{b}+\dfrac{k}{a}\right)y - (1+k) = 0$

$\therefore$ Slope = $-\dfrac{\left(\frac{1}{a}+\frac{k}{b}\right)}{\frac{1}{b}+\frac{k}{a}}$. But given that the slope is –1

$\therefore\ \dfrac{\left(\frac{1}{a}+\frac{k}{b}\right)}{\left(\frac{1}{b}+\frac{k}{a}\right)} = 1 \Rightarrow \dfrac{1}{a}+\dfrac{k}{b} = \dfrac{1}{b}+\dfrac{k}{a} \Rightarrow \dfrac{1}{a}-\dfrac{1}{b} = \dfrac{k}{a}-\dfrac{k}{b} \Rightarrow \left(\dfrac{1}{a}-\dfrac{1}{b}\right) = k\left(\dfrac{1}{a}-\dfrac{1}{b}\right) \Rightarrow k = 1$

23. Given $P(\alpha,\beta)$ lies on y = 6x – 1 $\Rightarrow \beta = 6\alpha - 1 \Rightarrow 6\alpha - \beta - 1 = 0$ .... (1)

Also $Q(\beta,\alpha)$ lies on 2x – 5y = 5 $\Rightarrow 2\beta - 5\alpha = 5 \Rightarrow 5\alpha - 2\beta + 5 = 0$ ......(2)

Solving (1) and (2), we have $\alpha = 1, \beta = 5$

$\therefore\ P = (1,5), Q = (5,1)$ Slope of PQ = $\dfrac{5-1}{1-5} = -1$

Equation of PQ is y – 5 = –1 (x – 1) $\Rightarrow$ y – 5 = – x + 1 $\Rightarrow x + y = 6$

Worksheet - 6 Date: 23-08-2010

1. Equation of the line is x $\cos\alpha$ + y $\sin\alpha$ = p. $\Rightarrow$ x $\cos 150^0$ + y $\sin 150^0$ = 6

$$\Rightarrow x\left(\frac{-\sqrt{3}}{2}\right)+y\left(\frac{1}{2}\right)=6 \Rightarrow -\sqrt{3}x+y=12 \Rightarrow \sqrt{3}x-y+12=0$$

2. Equation of the line is x cos 135^0 + y $\sin 135^0$ = 4 $\Rightarrow x\left(\frac{-1}{\sqrt{2}}\right)+y\left(\frac{1}{\sqrt{2}}\right)=4 \Rightarrow$ x – y + $4\sqrt{2}$ = 0

3. Given equation is 4x – 3y + 12 = 0

$$\Rightarrow \frac{-4}{\sqrt{4^2+3^2}}+\frac{3}{\sqrt{4^2+3^2}}y=\frac{12}{\sqrt{4^2+3^2}} \Rightarrow \frac{-4}{3}x+\frac{3}{5}y=\frac{12}{5}$$

4. Slope $=-\frac{1}{\sqrt{3}}=Tan\theta \Rightarrow \theta=150^0$ $\therefore$ Symmetric form is $\dfrac{x+2}{-\frac{\sqrt{3}}{2}}=\dfrac{y-0}{\frac{1}{2}}$

5. Key : 1, 2

6. Key : 2

Hint : Given equation is 4x - 3y + 12 = 0

$$\Rightarrow 4x-3y=-12 \Rightarrow \frac{-4}{5}x+\frac{3}{5}y=\frac{12}{5} \qquad \therefore p=\frac{12}{5}$$

7. Given equation is x - 3y+4=0

$\perp r$ from (0,0) to x-3y+4=0 is $\dfrac{|0+0+4|}{\sqrt{1+9}}=\dfrac{4}{\sqrt{10}}$

8. Given equation is $\dfrac{x}{a}+\dfrac{y}{b}=1$

bx + ay = ab

Normal form is $\dfrac{bx}{\sqrt{a^2+b^2}}+\dfrac{ay}{\sqrt{a^2+b^2}}=\dfrac{ab}{\sqrt{a^2+b^2}}$

$$\therefore p=\frac{ab}{\sqrt{a^2+b^2}} \Rightarrow \frac{1}{p^2}=\frac{a^2+b^2}{a^2b^2}=\frac{1}{a^2}+\frac{1}{b^2}$$

9. $$p=\frac{|a|}{\sqrt{\sec^2\alpha+\cos^2\alpha}}=\frac{|a|}{\sqrt{\frac{1}{\cos^2\alpha}+\frac{1}{\sin^2\alpha}}}=a\cos\alpha\cos\alpha=\left(\frac{a}{2}\right).\sin(2\alpha)$$

$$q=\frac{|a\cos^2\alpha|}{\sqrt{\cos^2\alpha\sin^2\alpha}}=a.\cos^2(\alpha)$$

$$\therefore 4p^2+q^2=4a^2\sin^2(2\alpha)+a^2\cos^2 2\alpha=a^2\left(\sin^2 2\alpha+\cos^2 2\alpha\right)=a^2$$

10. Key : $a\to q,\ b\to r,\ c\to s,\ d\to p$

a) Slope = $\sqrt{3} \Rightarrow \tan\theta = \sqrt{3} \Rightarrow \theta = 60° \Rightarrow \cos\theta = \frac{1}{2}, \sin\theta = \frac{\sqrt{3}}{2}$

symmetric form is $\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = \frac{x-2}{2} = \frac{y-3}{\sqrt{3}/2}$

b) Slope = $-1/\sqrt{3}, \tan\theta = -1/\sqrt{5} \Rightarrow \theta = 120° \Rightarrow \cos\theta = -1/2, \sin\theta = \frac{\sqrt{3}}{2}$

symmetric form is $\frac{x-1}{-1/2} = \frac{y-1}{\sqrt{3}/2}$

c) Slope = - 1 $\Rightarrow \tan\theta = -1 \Rightarrow \theta = 135° \Rightarrow \cos\theta = \frac{-1}{\sqrt{2}}, \sin\theta = 1/\sqrt{2}$

symmetric form is = $\frac{x-1}{-1/\sqrt{2}} = \frac{y-1}{1/\sqrt{2}}$

d) slope = - 1 $\Rightarrow \tan\theta = -1 \Rightarrow \theta = 135° \Rightarrow \cos\theta = \frac{-1}{\sqrt{2}}, \sin\theta = 1/\sqrt{2}$

symmetric form is = $\frac{x-2}{-1/\sqrt{2}} = \frac{y-3}{1/\sqrt{2}}$

11. Perpendicualr distance = $\frac{|-15|}{\sqrt{9+16}} = \frac{15}{5} = 3$

12. The parametric equation of the lines are $x = x_1 + r\cos\theta$, $y = y_1 + r\sin\theta$

$x = 3 + r\cos 135^0, y = 2 + r\sin 135^0 \Rightarrow x = 3 - \frac{r}{\sqrt{2}}, \quad y = 2 + \frac{r}{\sqrt{2}}$

13. Equation to the line is $\frac{x-4}{\cos 135^0} = \frac{y-1}{\sin 135^0} = r$

$x = 4 - \frac{r}{\sqrt{2}}, \quad y = 1 + \frac{r}{\sqrt{2}}$, Substituting in $3x - y = 0$

$\Rightarrow 12 - \frac{3r}{\sqrt{2}} - 1 - \frac{r}{\sqrt{2}} = 0 \quad \Rightarrow r = 11\sqrt{2}/4$

14. Key : 1,2

15. Let p be the length of the perpendicular from the origin on the given line. Then, its equation in normal form is $x\cos 30^0 + y\sin 30^0 = p \Rightarrow \sqrt{3}x + y = 2p$

$\Rightarrow \frac{x}{2p/\sqrt{3}} + \frac{y}{2p} = 1$

$\therefore$ Area of Δ OAB = $\frac{1}{2}\left(\frac{2p}{\sqrt{3}}\right)2p$ = $\frac{2p^2}{\sqrt{3}}$. But, Area of Δ OAB = $\frac{50}{\sqrt{3}}$ sq.units

$\therefore \frac{2p^2}{\sqrt{3}} = \frac{50}{\sqrt{3}} \Rightarrow$ p = ± 5.

Hence, the required lines are $\sqrt{3}x + y \pm 10 = 0$.

16. Given lines x cos α + y sin α = p_1 and x cos β + y sin β = p_2 will be perpendicular, if the lines perpendicular to them are also perpendicular.
Clearly, perpendiculars drawn from the origin to the given lines make angles α and β respectively with x-axis. Therefore, angle between them is $|\alpha - \beta|$.

Thus, the given lines will be perpendicular, if $|\alpha - \beta| = \frac{\pi}{2}$

17. Key: 1

18. The line is $\frac{x-3}{\cos 135^0} = \frac{y-4}{\sin 135^0} = \pm\sqrt{2}$ $\therefore$ x = 3 ∓ 1, y = 4 ± 1 $\Rightarrow$ (x, y) = (2, 5) and (4, 3)

19. Given $\tan\theta = 1 \Rightarrow \theta = 45^0$

Equation to the line is $\frac{x-1}{\cos 45} = \frac{y-1}{\sin 45} = r$

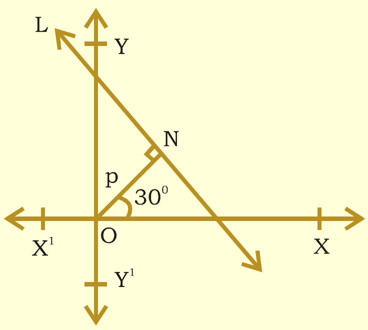
Given r = $5\sqrt{2}$. $\therefore \frac{x-1}{1/\sqrt{2}} = \frac{y-1}{1/\sqrt{2}} = 5\sqrt{2}$ $\Rightarrow$ (x, y) = (6, 6)

20. Key : 1
Hint : Let θ be the inclination of the line.

$\tan\theta = \frac{3}{4}$, equation to the line distance from is $\frac{x-2}{\cos\theta} = \frac{y-3}{\sin\theta} = r$

$x = 2 + r\cos\theta,\ y = 3 + r\sin\theta$ $\tan\theta = \frac{3}{4} \Rightarrow \cos\theta = \frac{4}{5}, \sin\theta = \frac{3}{5}$

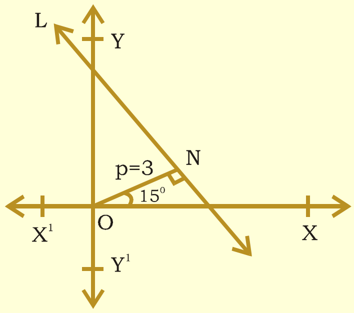
21. Here p = 5 units and $\alpha = 30^0$.
So, the equation of the given line in normal form is x cos α + y sin α = p, where $\alpha = 30^0$ and p = 5 units.



$\Rightarrow$ x cos30⁰ + y sin30⁰ = 5 $\Rightarrow x\left(\frac{\sqrt{3}}{2}\right) + y\left(\frac{1}{2}\right) = 5$

$\Rightarrow$ $\sqrt{3}$ x + y – 10 = 0, which is the required equation.

22. Here p = 3 units and $\alpha = 15^0$. So, the equation of the given line in normal form is $x\cos\alpha + y\sin\alpha = p$, where $\alpha = 15^0$ and p = 3 units.



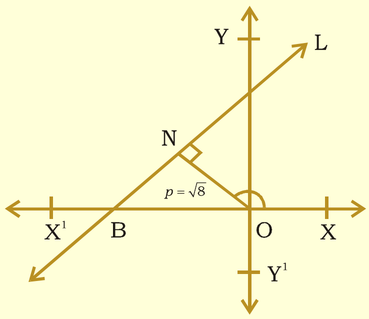
$\Rightarrow$ $x\cos 15^0 + y\sin 15^0 = 3 \Rightarrow \frac{x(\sqrt{3}+1)}{2\sqrt{2}} + \frac{y(\sqrt{3}-1)}{2\sqrt{2}} = 3$

[$\because$ $\cos 15^0 = \cos(45^0 - 30^0) = \cos 45^0 \cos 30^0 + \sin 45^0 \sin 30^0 = \frac{\sqrt{3}+1}{2\sqrt{2}}$

$\sin 15^0 = \sin(45^0 - 30^0) = \sin 45^0 \cos 30^0 - \cos 45^0 \sin 30^0 = \frac{\sqrt{3}-1}{2\sqrt{2}}$]

$\Rightarrow$ $(\sqrt{3}+1)x + (\sqrt{3}-1)y - 6\sqrt{2} = 0$, which is the required equation.

23. Here p = $\sqrt{8}$ units and α = 135^0. So, the equation of the given line in normal form is $x\cos\alpha + y\sin\alpha = p$, where $\alpha = 135^0$ and p = $\sqrt{8}$ units.



$\Rightarrow$ $x\cos 135^0 + y\sin 135^0 = \sqrt{8} \Rightarrow x\left(\frac{-1}{\sqrt{2}}\right) + y\left(\frac{1}{\sqrt{2}}\right) = \sqrt{8}$

[$\because$ $\cos 135^0 = \cos(180^0 - 45^0) = -\cos 45^0$, $\sin 135^0 = \sin(180^0 - 45^0) = \sin 45^0$]

$\Rightarrow$ $-x + y = 4 \Rightarrow x - y + 4 = 0$, which is the required equation.

Worksheet - 7 Date: 24-08-2010

1. Equation to the line is $\frac{x-3}{\cos\theta} = \frac{y-2}{\sin\theta} = r$, where $\tan\theta = \frac{3}{4}$

$\Rightarrow$ $\cos\theta = \frac{4}{5}$, $\sin\theta = \frac{3}{5}$, where $r = \pm 25$

The equation is $\frac{x-3}{(4/5)} = \frac{y-2}{(3/5)} = \pm 25 \Rightarrow \frac{x-3}{4} = \frac{y-2}{3} = \pm 5$

Then $x = 3 \pm 20$ and $y = 2 \pm 15$ $\Rightarrow$ (x, y) = (23, 17) and (–17, –13)

2. Let the line be $\frac{x-1}{\cos\theta}=\frac{y-1}{\sin\theta}=\sqrt{2}$

The point at a distance $\sqrt{2}$ is = $\left(1+\sqrt{2}\cos\theta,\ 1+\sqrt{2}\sin\theta\right)$

The point lies on x + y = 4 $\Rightarrow$ $1+\sqrt{2}\cos\theta+1+\sqrt{2}\sin\theta=4 \Rightarrow \cos\theta+\sin\theta=\sqrt{2}$

Squaring on both sides, we have $(\cos\theta+\sin\theta)^2=2 \Rightarrow 1+\sin 2\theta=2 \Rightarrow \sin 2\theta=1$

$\Rightarrow 2\theta=90^0 \Rightarrow \theta=45^0$

3. Equation to the line passing through (1, 1) and parallel to x + y = 1 is (x – 1) + (y – 1) = 0 $\Rightarrow$ x + y = 2.
Solving with 2x – 3y = 4, we get the point (2, 0)
Distance between (1, 1) and (2, 0) is $\sqrt{1+1}=\sqrt{2}$

4. We have $3x+4y-5=0 \Rightarrow 3x+4y=5$ Dividing on both sides by 5, we get

$\frac{3}{5}x+\frac{4}{5}y=1 \Rightarrow \frac{x}{\left(\frac{5}{3}\right)}+\frac{y}{\left(\frac{5}{4}\right)}=1$ which is the required intercept form.

Then intercept on x - axis is $\frac{5}{3}$ and the intercept on y - axis is $\frac{5}{4}$.

5. Key: 4

6. Key: [1] (Hint: Let θ be the inclination of the line

$\mathrm{Tan}\theta=\frac{3}{4}$, Equation to the line distance from is $\frac{x-2}{\cos\theta}=\frac{y-3}{\sin\theta}=r$

$x=2+r\cos\theta;\ y=3+r\sin\theta$

$\mathrm{Tan}\ \theta=\frac{3}{4} \Rightarrow \cos\theta=\frac{4}{5},\ \sin\theta=\frac{3}{5}$

7. Key: [4] $x+\sqrt{3}y+4=0 \Rightarrow \sqrt{3}y=-x-4 \Rightarrow y=\frac{-1}{\sqrt{3}}x+\frac{(-4)}{\sqrt{3}}$

8. Key: [2] $x+\sqrt{3}y=-4 \Rightarrow \frac{x}{-4}+\frac{\sqrt{3}y}{-4}=1 \Rightarrow \frac{x}{-4}+\frac{y}{\frac{-4}{\sqrt{3}}}=1$

9. Key: [2] $x+\sqrt{3}y+4=0 \Rightarrow x+\sqrt{3}y=-4 \Rightarrow -x-\sqrt{3}y=4$

Dividing both sides by $\sqrt{(-1)^2+\left(\sqrt{3}\right)^2}=\sqrt{4}=2$, we get $-\frac{1}{2}x-\frac{\sqrt{3}}{2}y=2$

10. a) $2x+3y+4=0,\ 2x+3y-5=0$

$a=2, b=3, c_1=4, c_2=-5$

$\therefore$ Distance = $\frac{|c_1-c_2|}{\sqrt{a^2+b^2}}=\frac{|4+5|}{\sqrt{4+9}}=\frac{9}{\sqrt{13}}$

b) $x-2y+3=0$ *and* $4y-2x+1=0$

$x-2y+3=0 \Rightarrow -2x+4y-6=0$

$\therefore a=-2, b=4,\ c_1=1,\ c_2=-6$

Distance = $\frac{|1+6|}{\sqrt{4+16}}=\frac{7}{\sqrt{20}}=\frac{7}{2\sqrt{5}}$

c) $x-y+5=0$ *and* $3x-3y-5=0$

$x-y+5=0 \Rightarrow 3x-3y+5=0$

$\therefore a=3,\ b=-3,\ c_1=15,\ c_2=-5$

Distance = $\frac{|15+5|}{\sqrt{9+9}}=\frac{20}{\sqrt{18}}=\frac{20}{3\sqrt{2}}$

d) $x+y+7=0$ and $x+y+8=0$

$a=1,\ b=1,\ c_1=7, c_2=8$

Distance = $\frac{|7-8|}{\sqrt{1+1}}=\frac{1}{\sqrt{2}}$

11. The symmetric form is $\frac{x-2}{\cos\theta}=\frac{y-1}{\sin\theta}=3\sqrt{2} \Rightarrow$ x = $2+3\sqrt{2}\cos\theta$, y = $1+3\sqrt{2}\sin\theta$

substituting in y – 2x + 6 = 0, we get $1+3\sqrt{2}\sin\theta-2\left(2+3\sqrt{2}\cos\theta\right)+6=0$

$\Rightarrow\ 2\sqrt{2}\cos\theta-\sqrt{2}\sin\theta=1 \Rightarrow 2\sqrt{2}-\sqrt{2}.\tan\theta=\sec\theta$

Squaring on both sides, we have $8-8\tan\theta+2\tan^2\theta=(1+\tan^2\theta)$

$\Rightarrow\ \tan^2\theta-8\tan\theta+7=0\ \Rightarrow (\tan\theta-1)(\tan\theta-7)=0 \Rightarrow\ \tan\theta=1\ \Rightarrow \theta=\frac{\pi}{4}$

12. Equation to the line passing through P(3, 4) and inclined at $\frac{\pi}{6}$ with x-axis is

$\frac{x-3}{\cos\frac{\pi}{6}}=\frac{y-4}{\sin\frac{\pi}{6}}=r\quad \Rightarrow x=3+\frac{r\sqrt{3}}{2},\ y=4+\frac{r}{2}$

Substituting in 3x + 5y + 1 = 0, we have $3\left(3+\frac{r\sqrt{3}}{2}\right)+5\left(4+\frac{r}{2}\right)+1=0$

$\Rightarrow\ r\left(\frac{3\sqrt{3}+5}{2}\right)+30=0\ \Rightarrow\ r=-30\left(\frac{2}{3\sqrt{3}+5}\right)=\frac{-30(2)\left(3\sqrt{3}-5\right)}{(27-25)}$

$\therefore$ Required PQ = $30(5-3\sqrt{3})$.

13. Slope $-\frac{1}{\sqrt{3}}=Tan\theta\ \Rightarrow\ \theta=120^0$

$\therefore$ Symmetric form = $\frac{x+2}{-1/2}=\frac{y-0}{\sqrt{3}/2}$

14. The parametric equation of the lines are x = 3 + r cos135^0, y = 2 + r sin135^0

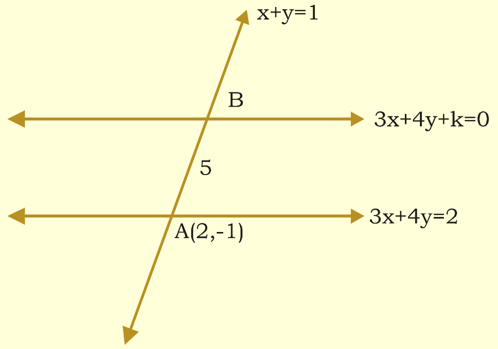
$\Rightarrow$ x = $3-\frac{r}{\sqrt{2}}$, y = $2+\frac{r}{\sqrt{2}}$

15. The line x + y = 1 makes on angle 135^0 with +ve x - axis
$\therefore$ the point B is 5 units away from (2, –1) on the line x + y = 1

B = (2 + 5 cos 135, –1 + 5 sin135) $=\left(2-\frac{5}{\sqrt{2}},-1+\frac{5}{\sqrt{2}}\right)$

Since B lies on 3x + 4y + k = 0

$\Rightarrow 3\left(2-\frac{5}{\sqrt{2}}\right)+\left(-1+\frac{5}{\sqrt{2}}\right)+k=0 \Rightarrow 6-\frac{15}{\sqrt{2}}-4+\frac{20}{\sqrt{2}}+k=0 \Rightarrow k=-2-\frac{5}{\sqrt{2}}$



The required equation is 3x + 4y – (2 + $\frac{5}{\sqrt{2}}$) = 0 $\Rightarrow 3x+4y=2+\frac{5}{\sqrt{2}}$

16. $\sqrt{3}\,x+y=4$ $\quad \frac{\sqrt{3}x}{2}+\frac{y}{2}=2 \Rightarrow x\cos\frac{\pi}{6}+y\sin\frac{\pi}{6}=2$

17. $4y=3x+6 \Rightarrow y=\left(\frac{3}{4}\right)x+\frac{3}{2}$

18. $4x+3y=9 \Rightarrow x/9/4+y/3=1$

19. $\frac{x-2}{\cos\theta}=\frac{y-1}{\sin\theta}=3\sqrt{2}$ $\Rightarrow$ x = $2+3\sqrt{2}\cos\theta$, y = $1+3\sqrt{2}\sin\theta$

substituting in y – 2x + 6 = 0, we get $1+3\sqrt{2}\sin\theta-2\left(2+3\sqrt{2}\cos\theta\right)+6=0$

$\Rightarrow$ $2\sqrt{2}\cos\theta-\sqrt{2}\sin\theta=1$ $\Rightarrow$ $2\sqrt{2}-\sqrt{2}.\tan\theta=\sec\theta$

squaring, 8 – 8 tanθ + 2 tan$^2\theta$ = (1 + tan$^2\theta$)

$\Rightarrow$ (tanθ – 1) (tanθ – 7) = 0 $\Rightarrow$ tanθ = 1 $\Rightarrow$ $\theta=\frac{\pi}{4}$

20. Equation to the line passing through P(3, 4) and inclined at $\frac{\pi}{6}$ with x-axis is

$\frac{x-3}{\cos\frac{\pi}{6}}=\frac{y-4}{\sin\frac{\pi}{6}}=r \Rightarrow x=3+\frac{r\sqrt{3}}{2},\ y=4+\frac{r}{2}$

Substituting in x + 5y + 1 = 0, we have $3\left(3+\frac{r\sqrt{3}}{2}\right)+5\left(4+\frac{r}{2}\right)+1=0$

$\Rightarrow\ r\left(\frac{3\sqrt{3}+5}{2}\right)+30=0\ \Rightarrow\ r = -30\left(\frac{2}{3\sqrt{3}+5}\right)=\frac{-30(2)\left(3\sqrt{3}-5\right)}{(27-25)}$

$\therefore$ Required PQ = $30(5-3\sqrt{3})$.

Worksheet - 8 Date: 25-08-2010

1. The slope of 2x + ky – 10 = 0 is $\frac{-2}{k}$

   and the slope of 5x + 2y –7 = 0 is $\frac{-5}{2}$

   The lines are parallel $\Rightarrow \frac{-2}{k}=\frac{-5}{2}\ \Rightarrow\ \frac{2}{5}=\frac{k}{2}\ \Rightarrow\ k=\frac{4}{5}$

2. Line perpendicular to 4x + 7y = 9 is 7x – 4y = λ and it passes through (2, 3). So $\lambda=2$.

3. Given lines are x – 3y + 1 = 0 ....(1); 2x + 5y – 9 = 0 .....(2)
   Solving (1) and (2), we have (x, y) = (2, 1)
   The required line is parallel to x – y + 2 = 0
   $\therefore$ y – 1 = 1(x – 2) $\Rightarrow$ x – y – 1 = 0 is the required line.

4. Slope of the line = 1 = tan45.
   $\therefore \theta=45°$ which is also the angle with the line y = 7.

5. Key : 1,3

   Hint : $\cos 45°=\left|\frac{4(k)+(-1)5}{\sqrt{16+1}\sqrt{k^2+25}}\right|\Rightarrow\frac{1}{\sqrt{2}}=\frac{|4k-5|}{\sqrt{17}\sqrt{k^2+25}}$

   $\Rightarrow 2(4k-5)^2=17\left(k^2+25\right)\Rightarrow 5k^2-8k-375=0\Rightarrow 3k^2-16k-75$

   $\Rightarrow (k+3)(3k-25)=0\Rightarrow k=-3,\ k=\frac{25}{3}$

6. Key : 1

7. Given lines are parallel $\Rightarrow \frac{2}{k}=\frac{3}{6}\Rightarrow k=4$

8. Given lines are perpendicular $\Rightarrow 3(2)+(-k)(1)=0\Rightarrow k=6$

9. Given lines are perpendicular $\Rightarrow 5(k)+3(-7)=0\Rightarrow k=\frac{21}{5}$

10. a) If θ is an angle then another angle = $\pi-\theta$

b) $\cos\alpha = \dfrac{|3\times1-3|}{\sqrt{9+1}\sqrt{1+9}} = 0$

c) $\cos\theta = \dfrac{\left|\dfrac{1}{ab}+\dfrac{1}{ab}\right|}{\sqrt{\dfrac{1}{a^2}+\dfrac{1}{b^2}}\sqrt{\dfrac{1}{b^2}+\dfrac{1}{a^2}}} = \dfrac{\dfrac{2}{ab}}{\dfrac{1}{a^2}+\dfrac{1}{b^2}} = \dfrac{2}{ab}\times\dfrac{ab}{\left(a^2+b^2\right)} = \dfrac{2ab}{a^2+b^2}$

d) $ax+by=a+b$

$$(a+b)x+(b-a)y=2b = \cos\theta = \left(\frac{a(a+b)+b(b-a)}{\sqrt{a^2+b^2}\sqrt{(a+b)^2+(b-a)^2}}\right)$$

$$\frac{\left(a^2+ab+b^2-ab\right)}{\sqrt{\left(a^2+b^2\right)}\sqrt{2.}\left(\sqrt{a^2+b^2}\right)} = \pm\frac{\left(a^2+b^2\right)1}{\left(a^2+b^2\right)\sqrt{2}}$$

$\cos\theta = \dfrac{1}{\sqrt{2}} \Rightarrow \theta = \dfrac{\pi}{4}$ another angle $= \pi - \dfrac{\pi}{4} = \dfrac{3\pi}{4}$

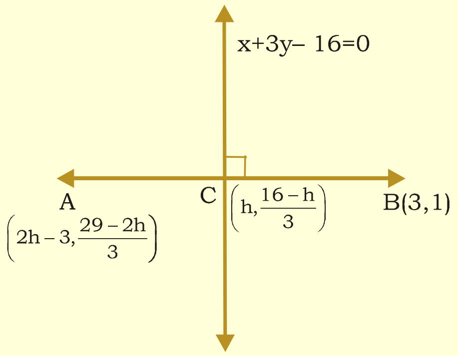
11. $\cos\dfrac{\pi}{4} = \dfrac{k(-3)+1}{\sqrt{k^2+1}.\sqrt{1+9}} \Rightarrow 10\left(k^2+1\right) = 2(1-3k)^2$

$\Rightarrow$ $2k^2 - 3k - 2 = 0$ $\Rightarrow$ $(2k + 1)(k - 2) = 0$ $\Rightarrow$ $k = -\dfrac{1}{2}, 2$

12. $\cos\theta = \dfrac{ab+ab}{\sqrt{a^2+b^2}\sqrt{a^2+b^2}} = \dfrac{2ab}{a^2+b^2}$

13. Key: 1

14. Let C(h, k) be the mid point of AB and it lies on $x + 3y - 16 = 0$



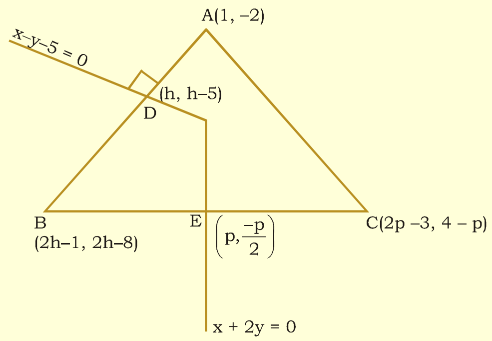
$\Rightarrow k = \dfrac{16-h}{3}$ $\quad\Rightarrow A = \left(2h-3, \dfrac{29-2h}{3}\right)$

Since x + 3y – 16 = 0 is the perpendicular bisector of AB $\Rightarrow \dfrac{\dfrac{29-2h}{3}-1}{2h-3-3}\times\dfrac{-1}{3}=-1$

$\Rightarrow$ 18h – 54 = 26 – 2h $\Rightarrow h = 4$

$\therefore$ A = (5, 7)

15. Given A = (1, –2); D = (h, h – 5) $\Rightarrow$ B = (2h –1, 2h – 8)



Since x – y – 5 = 0 is the perpendicular bisector of AB

$\Rightarrow \dfrac{2h-8+2}{2h-1-1}=-1 \Rightarrow$ h = 2 $\therefore B=(3,-4)$

Let $E=\left(p,\dfrac{-p}{2}\right) \Rightarrow C=(2p-3,4-p)$

Since x + 2y = 0 is the perpendicular bisector of BC $\Rightarrow \dfrac{-1}{2}\times\dfrac{4-p+4}{2p-3-3}=-1\Rightarrow p=4$

$\therefore$ C = (5, 0)

16. Here $m_1 = 2-\sqrt{3}$ and $m_2 = 2+\sqrt{3}$

If θ is the angle between lines, then $\tan\theta=\left|\dfrac{m_1-m_2}{1+m_1m_2}\right|$

$\Rightarrow \tan\theta=\left|\dfrac{2-\sqrt{3}-2-\sqrt{3}}{1+\left(2-\sqrt{3}\right)\left(2+\sqrt{3}\right)}\right| \Rightarrow \tan\theta=\dfrac{2\sqrt{3}}{1+(4-3)} \Rightarrow \tan\theta=\sqrt{3}$ $\therefore \theta=60°$

17. Equation to the perpendicular line is 2x + 3y = k.

Given, x - intercept = 3 $\Rightarrow \dfrac{k}{2}=3 \Rightarrow$ k = 6

$\therefore$ Equation to the line is 2x + 3y = 6 $\therefore$ intercept on y - axis is 2

18. The equation of a line passing through the point of intersection of $x-\sqrt{3}y+\sqrt{3}-1=0$ and x + y – 2 = 0 is $\left(x-\sqrt{3}y+\sqrt{3}-1\right)+\lambda(x+y-2)=0$

$\Rightarrow x(1+\lambda)+\lambda(x+y-2)=0$

$\Rightarrow x(1+\lambda)+y(\lambda-\sqrt{3})+\sqrt{3}-1-2\lambda=0$ --------- (1)

Now, the line $x-\sqrt{3}y+\sqrt{3}-1=0$ makes 30° angle with x-axis.
Since, the line makes an angle of 15° with this line.
Hence, the line will make 45° angle with x-axis.
$\therefore$ Its slope is 1

$$\Rightarrow -\left(\frac{1+\lambda}{\lambda-\sqrt{3}}\right)=1 \Rightarrow \lambda=\frac{\sqrt{3}-1}{2}$$

Putting the value of λ in (1), we get x – y = 0

19. Key: 1

20. Let (x_1, y_1) be the image of the point (1, 2) about the line y = x.

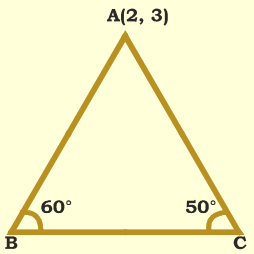
Then $x_1=2, y_1=1$ ....(1)

Also given image of $B(x_1, y_1)$ by the line mirror $y=0$ is (α,β).

Then $\alpha=x_1=2$ and $\beta=-y_1=-1$ $[\text{from}(1)]$

Hence, $\alpha=2$ and $\beta=1$

21. Let A(2, 3) be one vertex and x + y = 2 be the opposite side of an equilateral triangle. Clearly remaining two sides pass through the point A(2, 3) and make an angle of 60° with x + y = 2.

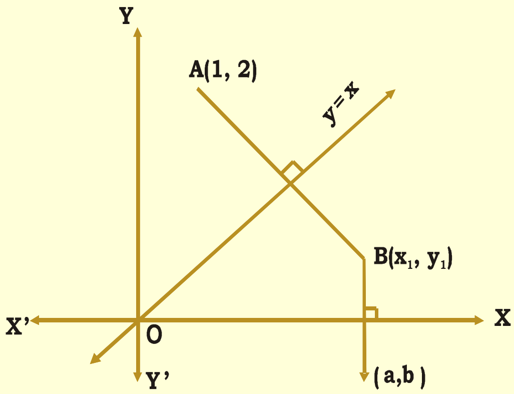


Let m be the slope of x + y = 2. Then, m = –1. So, the equation of the other sides are

$$y-3=\frac{-1-\tan 60°}{1+\text{tn}60°}(x-2) \text{ and, } y-3=\frac{-1+\tan 60°}{1+\tan 60°}(x-2)$$

$$\Rightarrow \quad y-3=\frac{-(1+\sqrt{3})}{1-\sqrt{3}}(x-2) \text{ and } y-3=\frac{\sqrt{3}+1}{\sqrt{3}+1}(x-2)$$

$$\Rightarrow \quad (2+\sqrt{3})x-y=1+2\sqrt{3} \text{ and } (2-\sqrt{3})x-y=1-2\sqrt{3}$$



Worksheet - 9 Date: 26-08-2010

1. Ratio = $-\left[\dfrac{3(2)-5(1)+7}{3(3)-5(-1)+7}\right]=\dfrac{-8}{21}$

   $\Rightarrow$ Points lie on same side of L = 0

2. Ratio = $-\left[\dfrac{3(2)+4(-5)-8}{3(-5)+4(2)-8}\right]=-\dfrac{22}{15}$

   $\Rightarrow$ Points lie on same side of L = 0

3. Given L = 2x + 3y – 5 = 0

   A(3, –2), B(1, 2) and C(1, –2)

   $-\dfrac{L_{(1,2)}}{L_{(1,-2)}}=-\left[\dfrac{2(1)+3(2)-5}{2(1)+3(-2)-5}\right]=\dfrac{1}{3}$; $-\dfrac{L_{(3,-2)}}{L_{(1,2)}}=-\left[\dfrac{2(3)+3(-2)-5}{2(1)+3(2)-5}\right]=\dfrac{5}{3}$

   $-\dfrac{L_{(3,-2)}}{L_{(1,-2)}}=-\left[\dfrac{2(3)+3(-2)-5}{2(1)+3(-2)-5}\right]=-\dfrac{5}{9}$

   $\Rightarrow$ A and C lie on the same side of L = 0.

4. Let the line perpendicular to 4x + 7y + 9 = 0 be 7x – 4y = k

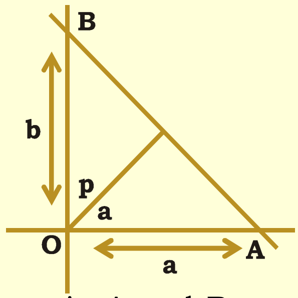
   $\Rightarrow \dfrac{x}{(k/7)}+\dfrac{y}{(-k/4)}=1$

   Given $\dfrac{1}{2}\left|\dfrac{k^2}{7(-4)}\right|$ = 3.5

   $\Rightarrow$ k^2 = 196 $\Rightarrow$ k = ±14

   Required line is 7x – 4y ±14 = 0.

5. Key: [2], [3]



Let the given line cut the axes in A and B
OA = a, OB = b;

Equation AB is $y - 0 = \frac{-b}{a}(x - a) \Rightarrow ay = -bx + ab \Rightarrow bx + ay - ab = 0$

$\therefore\ p = \frac{ab}{\sqrt{a^2 + b^2}} \Rightarrow p^2 = \frac{a^2b^2}{a^2 + b^2} \Rightarrow \frac{1}{p^2} = \frac{1}{a^2} + \frac{1}{b^2}$

6. Key: 4
7. Key: 1
8. Key: 2

Ratio = –(6 + 16 + 5) : (–9 – 4 + 5) = 27 : 8

9. Key: 2

$\overleftrightarrow{AB}$ equation is $y + 1 = \frac{1+1}{2+1}(x+1) \Rightarrow 3y + 3 = 2x + 2 \Rightarrow 2x - 3y - 1 = 0$

10. Key: 4

Equation of two line joining (2, 3), (4, –1) is $y - 3 = \frac{-1-3}{4-2}(x-2)$

$\Rightarrow (y-3) = -2(x-2) \Rightarrow 2x + y - 7 = 0$ Ratio $= -(2-2-7):(-6+2-7) = 7:-11$

11. Key: $a \to 3$; $b \to 2, 4, 5$; $c \to 2, 4, 5$; $d \to 2, 4, 5$

Hint: If L is line, then $\frac{-L_{11}}{L_{22}}$ in ratio.

a) $\frac{0+0+4}{2+2+4} = \frac{1}{2}$ b) $\frac{-2+2+4}{2+(-2)+4} = 1$ c) 1 d) 1

12. $\frac{x}{a} + \frac{y}{b} = 1$, $\frac{x}{p} + \frac{y}{q} = 1$ is same line in two different frames with same origin, distance of line in two different frames with same origin, distance of line from origin will remain same in both cartesian system.

$$\frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2}}} = \frac{1}{\sqrt{\frac{1}{p^2} + \frac{1}{q^2}}} \Rightarrow \frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{p^2} + \frac{1}{q^2}$$

13. $L_1(0,0) < 0,\ L_1(-2,1) < 0 \Rightarrow P_1$ and P_2 are on the same side of L_1.

Again $L_2(0,0) < 0, L_2(-2,1) < 0 \Rightarrow P_1$ and P_2 are on the same side of L_2.

$\Rightarrow$ P_1 and P_2 are contained in the same angle formed by L_1 and L_2 and therefore segment joining P_1 and P_2 will not cut L_1 and L_2 and therefore segment joining P_1 and P_2 will not cut L_1 or L_2.

14. Let the equation of the line be ax + by + c = 0.
Then, according to the given condition, we have

$$\frac{2a+0b+c}{\sqrt{a^2+b^2}}+\frac{0+2b+c}{\sqrt{a^2+b^2}}+\frac{a+b+c}{\sqrt{a^2+b^2}}=0 \Rightarrow 3a + 3b + 3c = 0 \Rightarrow a + b + c = 0$$

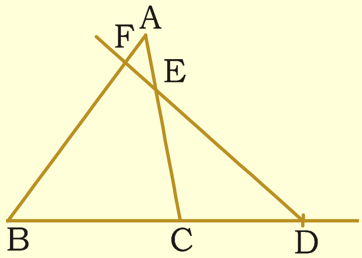
This shows that ax + by + c = 0 passes through the fixed point (1, 1)

15. The points are symmetrical about their perpendicular bisector.

$\therefore$ The required line is y = x.

16. Key: 1

Sol: Let the vertices of the triangle be $A(x_1,y_1)$, $B(x_2,y_2)$ and $C(x_3,y_3)$ and let the equation to the transversal $\overleftrightarrow{DF}$ be ax + by + c = 0



$$\frac{BD}{DC}=-\frac{ax_2+by_2+c}{ax_3+by_3+c}$$

$$\frac{CE}{EA}=-\frac{ax_3+by_3+c}{ax_1+by_1+c}$$

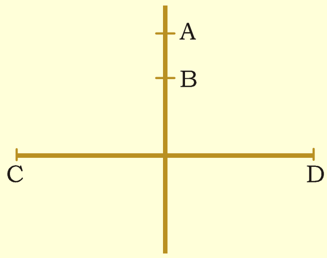
$$\frac{AF}{FB}=-\frac{ax_1+by_1+c}{ax_2+by_2+c}$$

$$\therefore \frac{BD}{DC}\times\frac{CE}{EA}\times\frac{AF}{FB}=-1.$$

17. Required ratio = $-(ax_1 + by_1 + c) : (ax_2 + by_3 + c)$

$= -[5(4) - 6(-1) - 21] : [5(2) - 6(1) - 21] = 5 : 17$

18. Slope of $\overleftrightarrow{AB}$ is $\frac{1+1}{2+1}=\frac{2}{3}$



Equation of $\overleftrightarrow{AB}$ is y + 1 = $\frac{2}{3}$(x + 1) $\Rightarrow$ 3y + 3 = 2x + 2 $\Rightarrow$ 2x – 3y – 1 = 0

The ratio in which $\overleftrightarrow{AB}$ divides $\overline{CD}$ = –[2(3) – 3(4) – 1] : [2(1) – 3(2) – 1]

= –(6 – 12 – 1) : (2 – 6 – 1) = –(–7) : (–5) = –7 : 5 i.e., 7 : 5 externally

19. Ratio = $-\left(\frac{3(0)+4(0)-8}{3(1)+4(1)-8}\right) = \frac{8}{-1}$ $\Rightarrow$ 8 : 1 externally

20. Key : 3, 4

The equation of side AB is $L(x,y) = \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} \Rightarrow L(0,0) = (x_1y_2 - x_2y_1)$

Also $L(x_3,y_3) = \begin{vmatrix} x_3 & y_3 & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = \begin{Vmatrix} lx_1+mx_2 & ly_1+my_2 & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{Vmatrix}$

$$= \begin{vmatrix} 0 & 0 & 1-l-m \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} (R_1 - lR_2 - mR_3) = (1-l-m)(x_1y_2 - x_2y_1)$$

$$\Rightarrow L(0,0)L(x_3,y_3) = (1--)(x_1y_2 - x_2y_1)^2 > 0$$

$\because$ origin and C lies on same side of AB.

Worksheet - 10 Date: 27-08-2010

1. The image of the line x = k in y - axes is x = -k.

2. $\frac{h-2}{4} = \frac{k-3}{-5} = \frac{-(4(2)-5(3)+8)}{4^2+5^2} \Rightarrow (h,k) = \left(\frac{78}{41}, \frac{128}{41}\right)$

3. If (h, k) is the foot of the perpendicular. $\Rightarrow \frac{h-x_1}{a} = \frac{k-y_1}{b} = \frac{-(ax_1+by_1+c)}{a^2+b^2}$

$$\Rightarrow \frac{h+3}{3}=\frac{k-7}{-4}=\frac{-[3(-3)-4(7)+12]}{(3)^2+(-4)^2}=1 \Rightarrow h=0, k=3$$

$\therefore$ The foot of the perpendicular is the point (0, 3)

4. $$\frac{h-0}{3}=\frac{k-0}{4}=\frac{-(3.0+4.0-25)}{3^2+4^2}=1$$

5. 1) $$\frac{h-3}{3}=\frac{k-4}{4}=-\frac{(9+16+25)}{25} \Rightarrow (h,k)=(-3,-4)$$

2) $$\frac{h-3}{3}=\frac{k-4}{4}=\frac{-2(9+16+25)}{25} \Rightarrow (h,k)=(-9,-12)$$

3) $$\left(\frac{3-9}{2},\frac{4-12}{2}\right)=(-3,-4)$$

4) equation $\overline{PQ},\ y-4=\frac{-4-4}{-3-3}(x-3) \Rightarrow 4x-3y=0$

6. $$\frac{h-6}{1}=\frac{k-1}{1}=\frac{-2(6+1)}{1+1} \Rightarrow (h,k)=(-1,-6)$$

In addition to the condition $\left(\frac{\alpha+\alpha^1}{2},\frac{\beta+\beta^1}{2}\right)$ lies on the gives line,

slope of $\overline{PQ}$ × slope of $ax+by+c=0$ is '-1' condition must be stated

so, statement 1 is correct, statement 2 is false .

7. Image of P (2,-3) w.r.t . x= y is $Q(-3,2)$

Image of $Q(-3,2)$ w.r.t (0,0) is R(3,-2)

8. Image of (4,6) w.r.t (3,2) is $(2(3)-4, 2(2)-6)$ = (2, -2)

9. As l = 2 ,m = 1,n = -1

tha equation of required line is $(x+y+1)(4+1)-2(2+1)(2x+y-1)=0$

$\Rightarrow 7x+y-11=0$

10. a) The reflection of (2,3) w.r.t (5,3) is (10-2,6-3) = (8,3)

b) $$\frac{h}{3}=\frac{k}{4}=\frac{25}{25} \Rightarrow (h,k)=(3,4)$$

c) The points are symmetrical about the $\perp r$ bisector

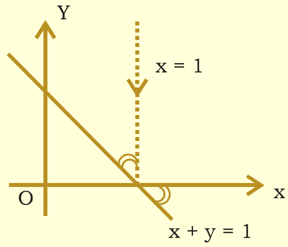
$\therefore$ the required line is y = x.

d) 'B' is the image of A(3,-3), w.r.t. x + y = 4. then B is (7,1)

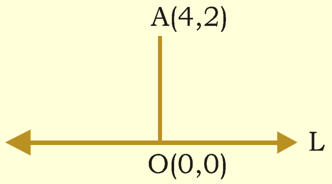
11. Reflected point through the origin = (–1, –2) = p.

$\therefore$ The reflection of p in x = y is (–2, –1)

12. Clearly from the figure the reflected ray moves along the x - axis
$\therefore$ Required equation is y = 0.

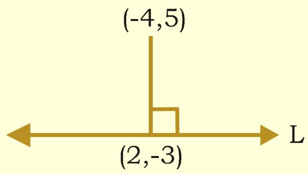


13. Key: 1
14. Slope of the line OA = 2/4 = 1/2. $\therefore$ Perpendicular slope of OA = –2



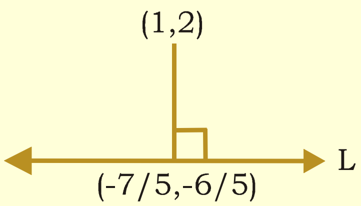
$\therefore$ Equation to the line is $y - 0 = -2(x - 0) \Rightarrow 2x + y = 0$.

15. Slope of the line perpendicular to L = $\dfrac{5+3}{-4-2} = \dfrac{-4}{3}$ $\therefore$ Slope of L = 3/4



Equation of the line L is $y + 3 = \dfrac{3}{4}(x - 2) \Rightarrow 3x - 4y = 18$

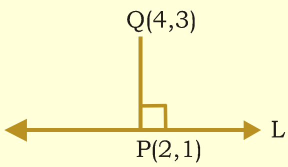
16. Slope of the line perpendicular to L = $\dfrac{2+(6/5)}{1+(7/5)} = \dfrac{4}{3}$ $\therefore$ Slope of L = – 3/4



Equation to the line L is $y + \dfrac{6}{5} = -\dfrac{3}{4}\left(x + \dfrac{7}{5}\right) \Rightarrow 3x + 4y + 9 = 0$.

17. Key: 1

18. Slope of PQ = $\dfrac{2}{2} = 1$, required line is passing through P (2, 1)



Required line $(y - 1) = -1(x - 2) \Rightarrow x + y = 3$.

19. Let (α,β) be the image of (–4, 3) $\Rightarrow \frac{h-x_1}{a}=\frac{k-y_1}{b}=\frac{-2(ax_1+by_1+c)}{a^2+b^2}$

$$\Rightarrow \frac{\alpha+4}{7}=\frac{\beta-3}{-24}=\frac{-2[7(-4)-24(3)+50]}{7^2+(-24)^2}=\frac{4}{25} \quad \therefore \quad \alpha=-\frac{72}{25},\ \beta=\frac{-21}{25}$$

$\therefore$ Image = $\left(\frac{-72}{25},\frac{-21}{25}\right)$

20. Given line is perpendicular bisector of AB $\Rightarrow$ B is the image of A with respective the line x + y – 4 = 0. Let B = (h, k)

$$\therefore \frac{h-3}{1}=\frac{k+3}{1}=\frac{-2(3-3-4)}{1+1}=4 \quad \Rightarrow (h, k) = (7, 1)$$

21. $a\rightarrow 3,\ b\rightarrow 4,\ c\rightarrow 1, d\rightarrow 2$

22. The image of the point A(1, 2) with respect to y = x is B(2, 1).

    The image of B(2, 1) with respect to y = 0 is the point (2, –1).

23. That line is perpendicular bisector to P(2, 1) and Q(4, 3) $\Rightarrow$ x + y = 5 is the required line.

24. The image of the point (4, 3) about the line

$$y = x \Rightarrow \frac{h-4}{1}=\frac{k-3}{-1}=\frac{-2(4-3)}{2} \Rightarrow (h, k)=(3, 4)$$

25. The slope of a line equally inclined with the co-ordinate axes is ± 1. So, let its equation be $x \pm y + \lambda = 0$

CASE : I

When $x + y + \lambda = 0$ is the line. It is equidistant from (1, –2) and 3, 4).

$$\therefore \left|\frac{1-2+\lambda}{\sqrt{2}}\right|=\left|\frac{3+4+\lambda}{\sqrt{2}}\right| \Rightarrow |\lambda-1|=|\lambda+7| \Rightarrow \lambda-1=-(\lambda+7) \Rightarrow \lambda=-3$$

So, the required line is x + y – 3 = 0.

CASE : II

When $x - y + \lambda = 0$ is the line. It is equidistant from (1, –2) and (3, 4)

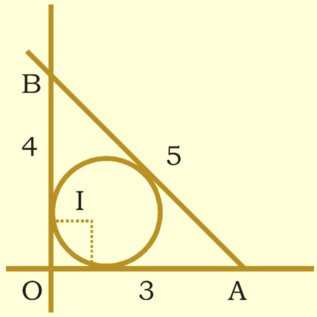
$$\therefore \left|\frac{1+2+\lambda}{\sqrt{2}}\right|=\left|\frac{3-4+\lambda}{\sqrt{2}}\right| \Rightarrow |\lambda+3|=|\lambda-1| \Rightarrow \lambda+3=-(\lambda-1) \Rightarrow \lambda=-1$$

Thus, the required line is x – y – 1= 0

Worksheet - 11, 12 Date: 30 & 31-08-2010

1. Key: 2
2. OA = 3, OB = 4 $\Rightarrow$ AB = 5

$\therefore$ 2s = 3 + 4 + 5 = 12 $\Rightarrow$ s = 6



r = Area Δ OAB = $\frac{1}{2}(3)\ (4) = 6$.

$\therefore$ r = $\frac{\Delta}{s} = \frac{6}{6} = 1$.

$\therefore$ Incentre = (1, 1)

3. Since x – y = 0 is perpendicular to x + y + 2 = 0, their intersection point (–1, –1) is the orthocentre.

4. Slope of the join of B(5, –1) and C(–2, 3) is $\frac{3+1}{-2-5} = \frac{-4}{7}$

$\therefore$ Slope of OA = $\frac{7}{4}$ $\Rightarrow$ A = (–4, –7)

5. Let $S(x,y)$ be the circumcentre

$SA = SB = SC \Rightarrow SA^2 = SB^2 = SC^2$

$$\Rightarrow (x+2)^2 + (y-3)^2 = (x-2)^2 + (y+1)^2 = (x-4)^2 + (y-0)^2 \Rightarrow (x,y) = \left(\frac{3}{2},\frac{5}{2}\right).$$

6. Key: 2. Hint: $\begin{vmatrix} 3 & 1 & 2 \\ 2 & -1 & 3 \\ a^2 & 2a & 6 \end{vmatrix} = 0$

7. Key: 1

8. Third vertex = $\left(\frac{x_1 + x_2 \pm \sqrt{3}\left(y_1 - y_2\right)}{2}, \frac{y_1 + y_2 \mp \sqrt{3}\left(x_1 - x_2\right)}{2}\right)$

$$= \left[\frac{2+2\pm\sqrt{3}(4-6)}{2}, \frac{(4+6)\mp\sqrt{3}(0)}{2}\right] = \left(\frac{4\pm\sqrt{3}(-2)}{2}, \frac{10}{2}\right)$$

$$= \left(\frac{4\mp2\sqrt{3}}{2}, 5\right) = \left(2\mp\sqrt{3}, 5\right) = \left[2--\sqrt{3}, 5\right] \; and \; \left(2+\sqrt{3}, 5\right)$$

9. orthocentre = centroid = $\left(\frac{6-\sqrt{3}}{3}, 5\right)$ or $\left(\frac{6+\sqrt{3}}{3}, 5\right)$

10. As the circumcentre and orthocentre coincides ,distance between them is ‘ 0 ’

11. $a\to4, b\to3, c\to2, d\to1$

12. $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0 \Rightarrow a^3 + b^3 + c^3 - 3abc = 0 \Rightarrow (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = 0$

$\Rightarrow \frac{a+b+c}{2} \{(a - b)^2 + (b - c)^2 + (c - a)^2\} = 0 \Rightarrow a + b + c = 0$

$(\because a \neq b \neq c)$

13. (a,b,c) for concurrency of three lines $px + qy + r = 0; qx + ry + p = 0; rx + py + q = 0$

we must have, $\begin{vmatrix} p & q & r \\ q & r & p \\ r & p & q \end{vmatrix} = 0$

Operate $C_1 + C_2 + C_3 \begin{vmatrix} p+q+r & q & r \\ p+q+r & r & p \\ p+q+r & p & q \end{vmatrix} = 0 \Rightarrow (p+q+r)\begin{vmatrix} 1 & q & r \\ 1 & r & p \\ 1 & p & q \end{vmatrix} = 0$

Operate $C_1 - C_2 - C_3 \Rightarrow (p+q+r)\begin{vmatrix} 0 & q-r & r-p \\ 0 & r-p & p-q \\ 1 & p & q \end{vmatrix} = 0$

$\Rightarrow (p+q+r)(pq - q^2 - rp + rq - r^2 + pr + pr - p^2) = 0$

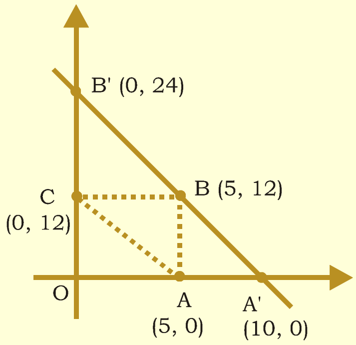
$\Rightarrow (p+q+r)(p^2 + q^2 + r^2 - pq - pr - rp) = 0 \Rightarrow p^3 + q^3 + r^3 - 3pqr$

It is clear that a,b,c are correct options.

14. Solving $x - 2y = 6$ and $3x + y = 4$, we get $(x, y) = (2, -2)$

Now. $2\lambda - 8 + \lambda^2 = 0 \Rightarrow (\lambda+4)(\lambda-2) = 0 \Rightarrow \lambda = 2, -4$

15. $AB \perp BC$.



The vertices of the triangle are O(0, 0), A'(10, 0), B'(0, 24).

$\therefore$ Orthocentre of ΔAOB is O(0, 0)..

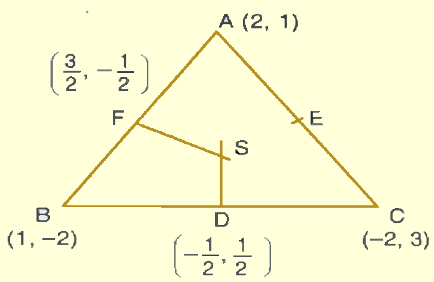
16. In any triangle, the centroid and the incentre lie within the triangle. If the triangle is obtuse angled, then the orthocentre and the circumcentre fell outside the triangle. Here, the angle between 7 y = x and $\sqrt{3}y + x = 0$ is obtuse because the first line is inclined at an angle less than 45° with the x - axis, while he second line makes 150° with the x - axis.

$\therefore$ (2), (3) hold.

17. The three lines are concurrent if $\begin{vmatrix} 1 & 2 & -9 \\ 3 & 5 & -5 \\ a & b & -1 \end{vmatrix} = 0$

$\Rightarrow$ 35a – 22b + 1 = 0 $\Rightarrow$ 35x – 22y + 1 = 0 passes through (a, b)

18.



Let A (2, 1), B (1, - 2), C (- 2, 3)

Mid point of BC = D = $\left(\frac{1-2}{2}, \frac{-2+3}{2}\right) = \left(-\frac{1}{2}, \frac{1}{2}\right)$

Mid point of AB = F = $\left(\frac{2+1}{2}, \frac{1-2}{2}\right) = \left(\frac{3}{2}, -\frac{1}{2}\right)$

Slope of BC = $\frac{-2-3}{1+2} = -\frac{5}{3} \Rightarrow$ Slope of SD = $\frac{3}{5}$

$\therefore$ Equation to the perpendicular bisector $\overleftrightarrow{SD}$ of $\overline{BC}$ is

$$y-\frac{1}{2}=\frac{3}{5}\left(x+\frac{1}{2}\right)\Rightarrow 6x-10y+8=0 \qquad \text{.............. (1)}$$

Slope of AB = $\frac{1+2}{2-1}=3$ $\therefore$ Slop of $\overleftrightarrow{SF}=-\frac{1}{3}$

Equation to the perpendicular bisector of is

Solving (1) and (2) : Circumcentre of ABC is

19. The ΔPQR form a right angled triangle, right anhgled at P(0,0) and QR is hypotenuse. So P(0,0) is orthocentre and the mid. point S of AR, i.e., (1,1) is circumcentre .

20. Given line a (x – y – 1) + b(2x – 3y + 1) = 0 passes through the inter section of x + y = 1 and 2x – 3y + 1 =0 $\therefore$ Solving the point is (2/5, 3/5]

21 to 23.

Here the coordinates of the vertices are (0,0), (2,0) and $\left(1,\sqrt{3}\right)$ which is an equilaterla triangle, and in any equilateral triangle all the three reqd. points coincide and will be $\left(1,\frac{1}{\sqrt{3}}\right)$

$\therefore$ (4) holds. for (22) (1) for (23) and (4) for (24)